

3-D Mesh with 2-D Convection

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MENG 451 Final Project

1-D Convection

- Apply 1-D Convection Boundary condition to Problem 2.
- Have to create another part of the mesh with line elements.
- Have to apply the boundary condition to stiffness matrix.

$$K_e = \int_{\Omega} h N^T N \det(J) d\xi$$

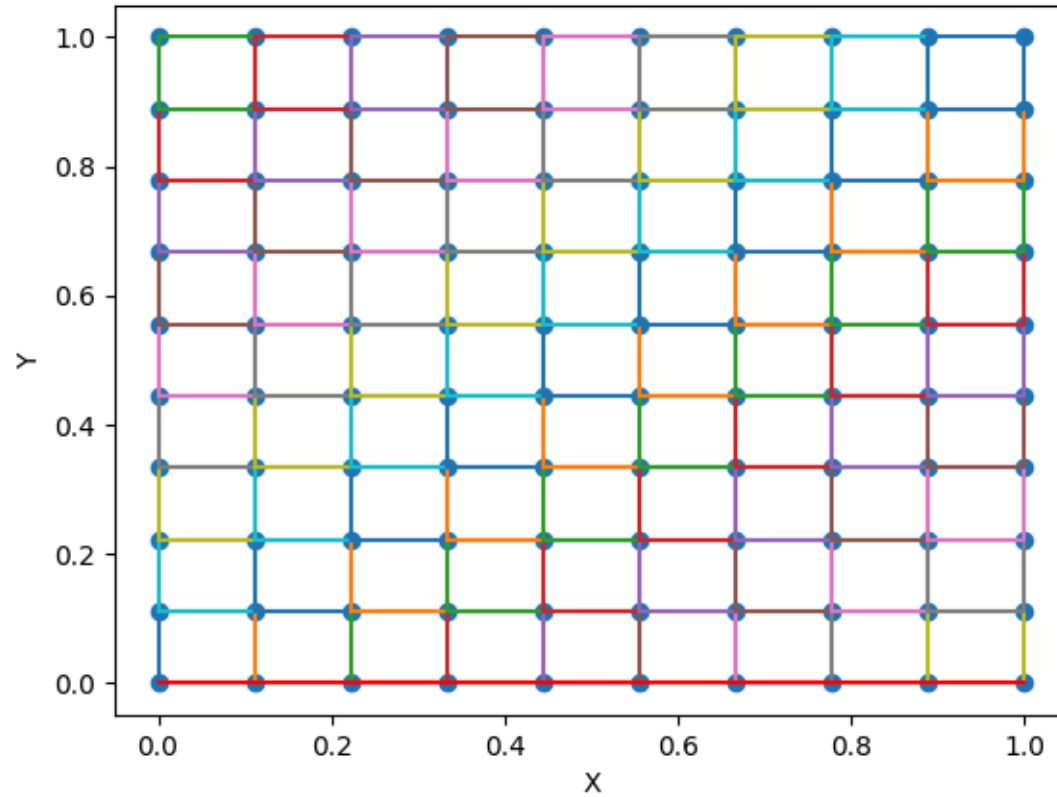


Figure 1: 2-D Mesh with 1-D Convection.

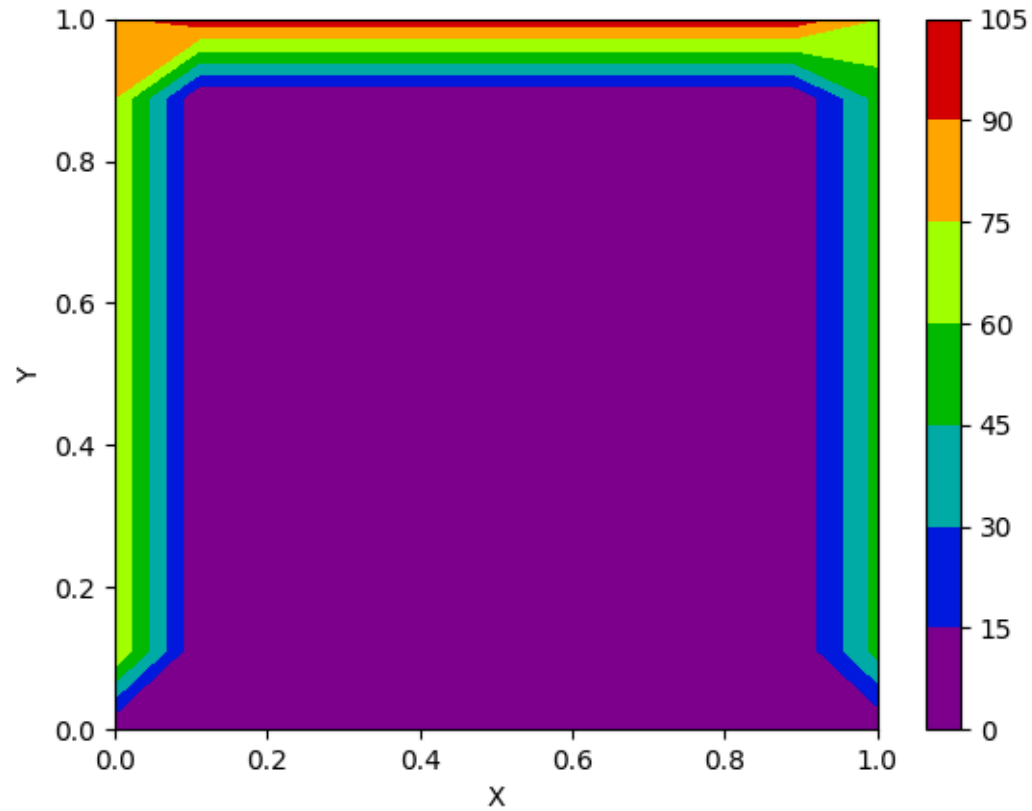


Figure 2: 2-D Boundary Conditions

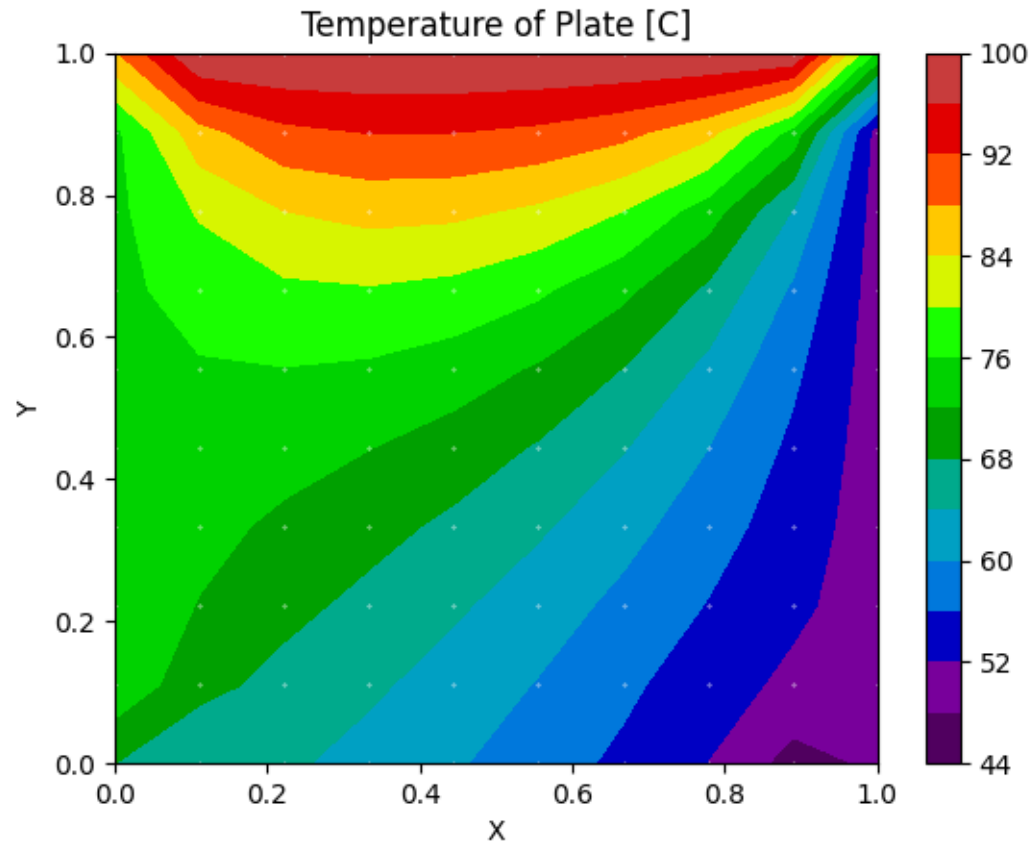


Figure 3: 2-D Results

3-D Mesh

- Must convert K_e to handle 3 dimensions.
- Have to set up mesh correctly.
- Add shape function for hexahedron.

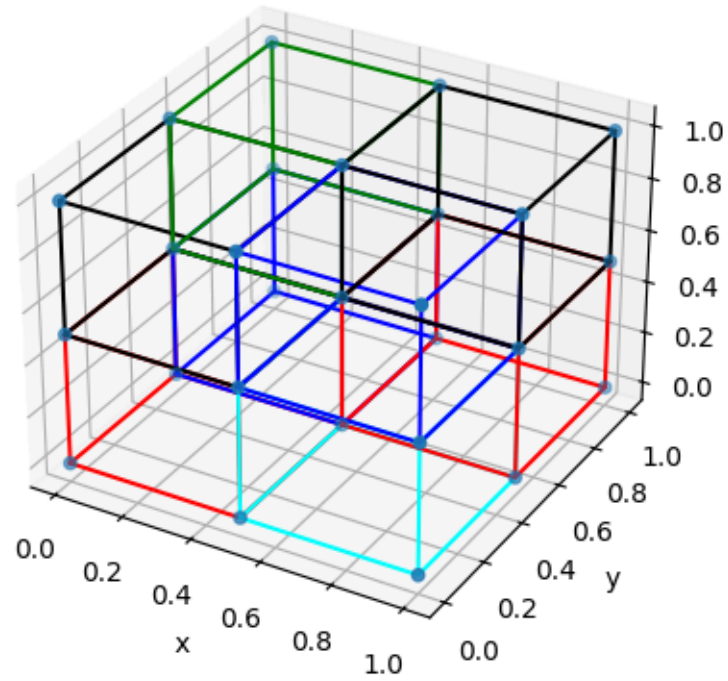


Figure 4: 3-D Mesh

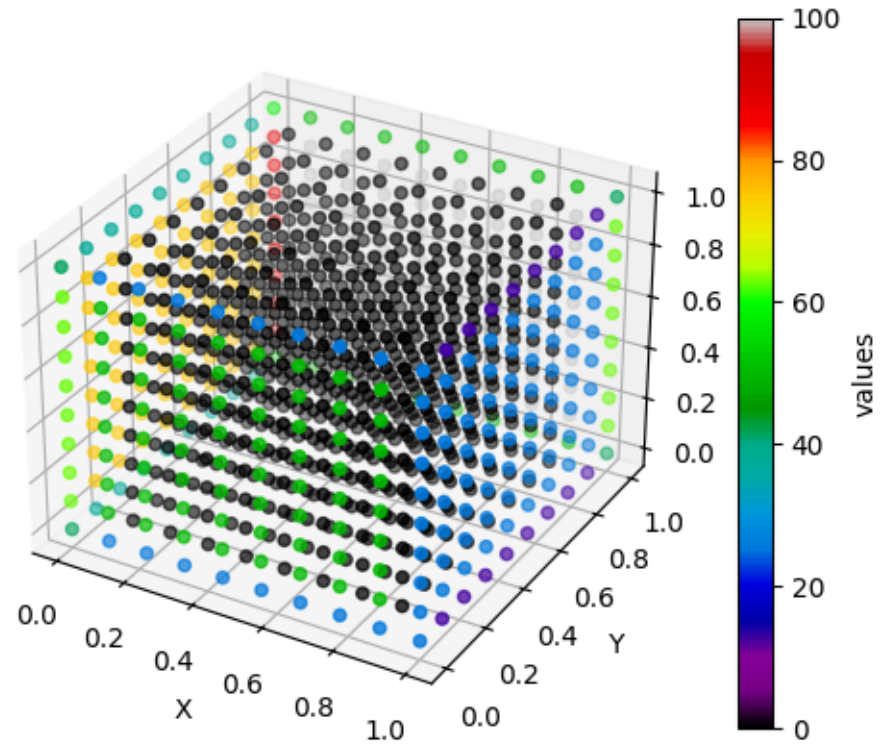


Figure 5: 3-D Boundary Conditions

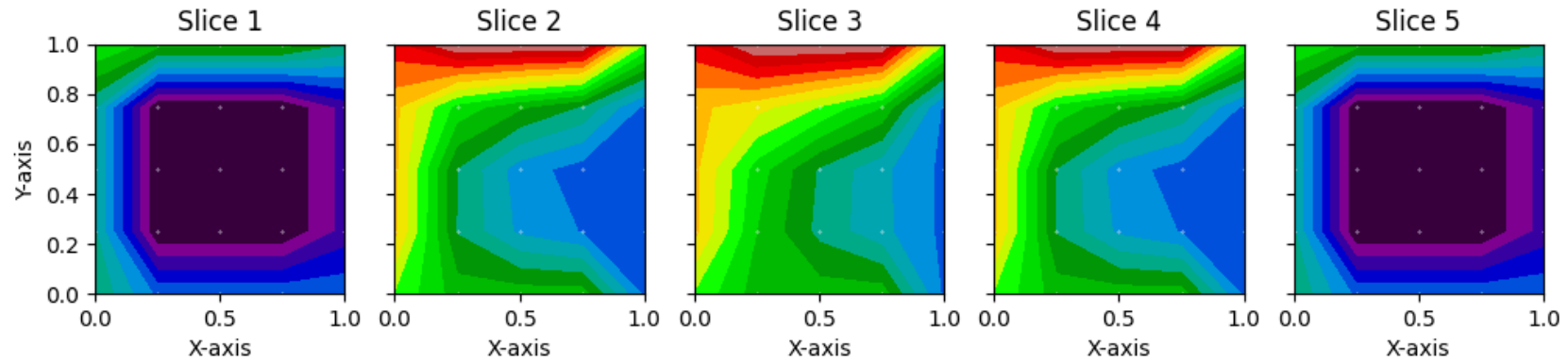


Figure 6: 3-D Results

2-D Convection

- Remove appropriate boundary condition.
- Change convection from line to quadrilateral elements.
- Change the modification of K_e to work with quadrilateral elements.

$$K_e = \int_{\Omega} h N^T N \det(J) d\xi d\eta$$

$$\det(J) = \|\overrightarrow{r}_{,\xi} \times \overrightarrow{r}_{,\eta}\|$$

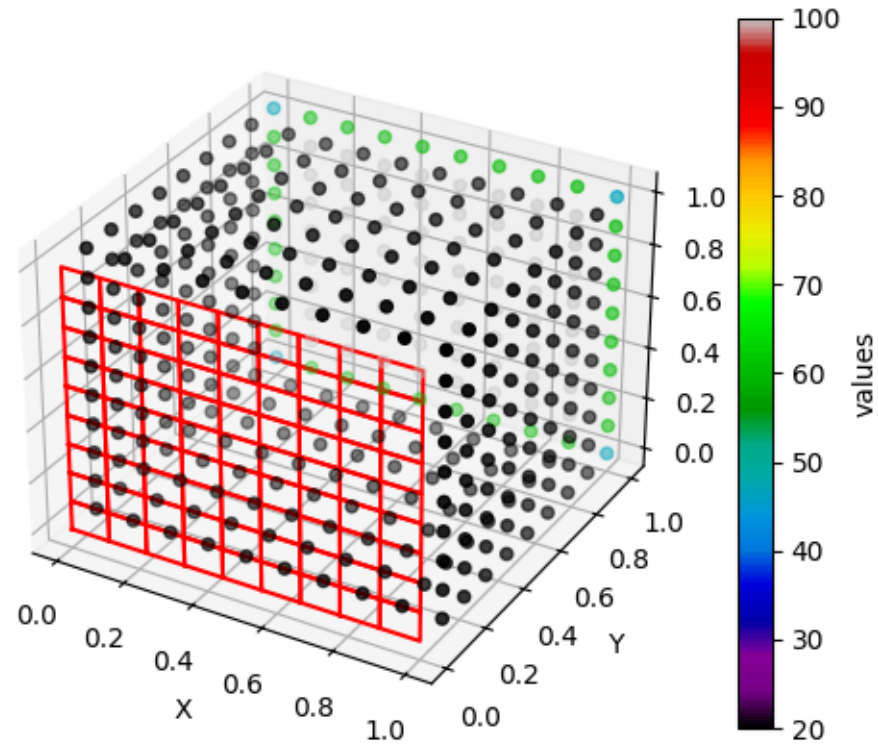


Figure 7: 3-D Convection Boundary Conditions

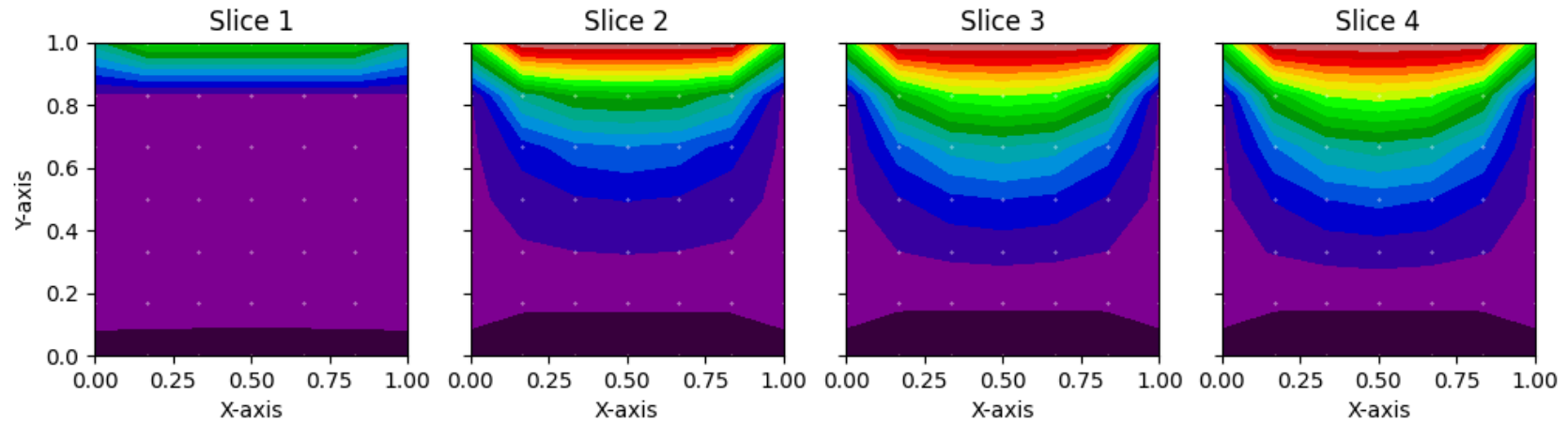


Figure 8: 3-D Convection Results